Technology and Health Care

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Journal:	*Technology and Health Care*
Manuscript ID	Draft
Manuscript Type:	Original Research Article
Date Submitted by the Author:	n/a
Complete List of Authors:	YAMAMOTO, SHUJI; LISIT Co., Ltd., Institute of One
Keywords:	Computed tomography, Cardiac imaging techniques, Image processing, computer-assisted, Signal processing, computer-assisted, Sampling studies, Reproducibility of results
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SCHOLARONE MANUSCRIPTS

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Abstract

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METHODS: The cycles needed were measured over 144 simulated conditions, applied to 192 real series by header, and tested on 17 image series under a frozen criterion.

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CONCLUSIONS: Whether a scan can carry cardiac timing follows from its header, before any image is analysed. A period estimated from images should not be accepted on self-consistency alone: no level of the admitted series tracked the intended border throughout, and re-counting over the stretch where it did put one of the three below the cycle threshold. Before clinical use a period needs the depth of its minimum and a check of what was tracked.

Keywords

Computed tomography; Cardiac imaging techniques; Image processing, computer-assisted; Signal processing, computer-assisted; Sampling studies; Reproducibility of results

1. Introduction

A coronal or sagittal reformat of a non-gated helical chest CT shows the left mediastinal border as an undulating line. It is usually read as a motion artefact. Part of it is written by the heart: the table advances while the heart moves, so cardiac motion contributes to the shape of that line whether or not an electrocardiogram was connected.

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That is what makes the question in this paper worth asking, and the step from it to a use is a hypothesis rather than a result. If a component attributable to cardiac motion can be separated from the rest of the line, then the spacing of its successive maxima would be a cue to the cardiac period, and its maxima and minima, falling on opposite parts of the cycle, could serve as cues to phase; a scan already acquired might then be sorted by phase and reformatted accordingly, and further along that road used for ventricular volumetry. Such a route would need no electrocardiographic hardware, no additional acquisition and no repeated overlapping rotations over the heart. This paper examines one point on that route — whether the period can be recovered — and examines nothing beyond it. Estimating phase without first estimating a period is a different route and is not studied here either. Phase identification, phase-resolved reconstruction, volumetry and ejection fraction are not studied, and nothing below should be read as evidence about them.

The opportunity is large because the examinations already exist. Chest CT is acquired in very large numbers, and lung cancer screening alone has made it a population-scale examination;[1] the heart is in every one of those scans. Automated methods increasingly read it: fully automated deep-learning coronary calcium scoring now runs on non-gated low-dose chest CT,[2] and its output depends on the acquisition parameters of the scan it is given.[3] In the gated setting, deep learning is also used to remove cardiac motion from the images themselves.[4]

A method of that kind makes a claim about a heart that was never synchronised to the scanner. The obvious way to check such a claim is to compare it with the rate the scanner recorded, and that comparison is not available. Of the 18 series whose images I analysed, none records the rate by any of 11 routes through which a CT acquisition can write one.

Rather than assess any particular method, I ask what the acquisition can contain. Recovering timing from the image itself is not a new idea in CT. Reconstruction correlated with a recorded electrocardiogram came first,[5] and was then replaced by a signal read out of the projection data — the kymogram[6] — and later by gating estimated from the reconstructed images themselves.[7] The same idea carried small-animal imaging, where respiratory and cardiac gating without a physiological monitor became routine.[8,9] Every one of those settings has an acquisition that dwells: a cardiac or micro-CT protocol writes many cycles into the data. That line of work largely stops around 2009.

A separate literature treats the heart in non-gated chest CT, but structurally — thrombus, calcification, aortic root, incidental infarct, epicardial fat. Gating inferred from the data rather than from a monitor has continued elsewhere, in MRI[10] and in PET, where it now outperforms device-based gating,[11] but not in helical CT of the chest. I found no indexed report that recovers cardiac timing from a non-gated helical CT, and none that states the condition under which it could be recovered.

This paper sets out the two sampling requirements recovery has to satisfy, shows that their product is fixed by anatomy and reconstruction interval alone, and measures the constant one of them needs for

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one estimator on simulated traces. It then evaluates the requirements against the installed base from acquisition headers, and tests recovery on images under a criterion frozen in advance. That image test is exploratory: no public acquisition I could find pairs a reconstructed helical chest series with a recorded rate, so a self-consistent estimate cannot be confirmed as a correct one, and I report which further measurements separate the two. A simulation in the supplementary material asks what uncertainty a learned estimator declares outside its training range. The contribution is a way of connecting acquisition parameters to the interpretation of an estimate: a framework for choosing the acquisitions on which a future period or phase estimator could be tested, and for reading what its output means. It is not a limit on the cardiac information a CT can hold, and it is not a demonstration of clinical usefulness.

2. Theory

2.1 The z axis of a helical scan is a time axis

In a helical acquisition the table advances at a constant speed S while the gantry rotates, so the axial coordinate of the reconstructed volume and the time of acquisition are related by $z = St$. The scale is known from the header, and the *nominal* acquisition time it gives is the same for every voxel at a given z whatever its row or column: to that approximation, a slice is a slice of time as much as of anatomy. How far the reconstructed value at a voxel departs from that nominal instant is the subject of the assumptions below.

A structure that moves periodically with period T therefore writes that period into the volume as a *spatial* period along z,

$$\lambda = S\,T.$$

For the heart this is the wavy left mediastinal silhouette familiar from a coronal reformat. Below the heart the same silhouette continues as the descending aorta against the left lung, extending the length over which the wave is available.

Two assumptions are carried by $\lambda = ST$ and neither is tested in this study. The first is that the table speed is constant across the analysed span and that the header reports it correctly. The second concerns reconstruction. Each reconstructed slice is formed from projections spanning a finite gantry interval and, on a multi-row detector, from interpolation along z, so the border position recorded at a given z is an average over a temporal footprint rather than an instantaneous position. Such averaging attenuates the periodic component and can displace its apparent phase, by an amount that depends on the reconstruction algorithm, the collimation and the interpolation used. The relation $z = St$ describes the acquisition; it is not a guarantee that the periodic displacement reaches the voxels undistorted, and the simulations of section 3 model the trace rather than the reconstruction that produced it.

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2.2 The recoverability window

Whether λ can be recovered is a sampling question with two sides that pull against each other.

The scan must be slow enough that more than one cycle is written across the structure. If the available craniocaudal extent is L, the number of cycles written is $N = L/\lambda = L/(ST)$, and measuring a period requires N to exceed some Nmin. This places an upper limit on S.

The scan must be fast enough that each cycle is spread across enough reconstructed samples to be resolved. With reconstruction interval dz, the samples per cycle are $n = \lambda/dz = ST/dz$, and this must exceed some $n$min. This places a lower limit on S.

The value of $n$min is a judgement and is the one such number in this analysis. Two samples per cycle is the formal floor and is not usable, because a two-sample sinusoid is a straight line to any estimator with noise in it. I take eight throughout, and the sweep of section 3 covers 8, 16 and 32 samples per cycle so that the effect of the choice on Nmin can be read off rather than assumed. Nothing in the budget identity below depends on it.

Together they define a window in table speed, in closed form, for a given L, dz and heart rate. The trade-off itself is old — pitch has always been chosen against longitudinal sampling[12,13] — but the quantity being traded here is cardiac timing rather than spatial resolution. The acquisition side of the window comes entirely from the DICOM header of any helical scan: table speed (0018,9309), or equivalently spiral pitch factor (0018,9311) with total collimation (0018,9307) and revolution time (0018,9305). The other two quantities are not in the header. L is an assumption about which structure carries the motion, and T is the patient's period, which none of the series studied here records. The window is therefore evaluated under stated assumptions about both, which is why section 4 reports it at two values of L.

One ambiguity has to be closed explicitly. The spiral pitch factor of the IEC convention is table travel per rotation divided by the *total* collimated width, whereas the convention in use on four-row scanners divided by a *single* row width; the two differ by the number of rows. The implementation converts between them rather than accepting either silently.

2.3 The budget identity

Multiplying the two quantities removes the heart rate and the table speed at once:

$$N \times n = (L/ST) \times (ST/dz) = L / dz.$$

The product of cycles observed and samples per cycle is fixed by the anatomy and the reconstruction interval alone. It does not depend on pitch, on rotation time, or on how fast the heart is beating. A protocol cannot acquire more cardiac timing by running faster or slower; it can only divide a fixed

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budget between seeing more cycles and resolving each one better. Making a scan slower buys cycles at the cost of resolution within a cycle, and making it faster does the reverse.

This is why the question has a definite answer for each protocol, and why the answer is computable before any image exists (figure 1). It also locates the one quantity the condition still needs: the window is bounded by *N*min, and *N*min is not given by the geometry. Section 3 measures it.

3. How many cycles are needed

3.1 Simulation design

How precisely a frequency can be estimated from a finite record is a classical question, and the answer depends on the observation length, the signal-to-noise ratio and the estimator.[14] What that theory does not give is the number of cycles at which a particular estimator, reading a particular signal, starts to succeed in practice. *N*min is a property of the estimation problem — of noise, of how well the cardiac border can be located against lung and mediastinum, and of how far a real rhythm departs from a fixed period — so it was measured rather than assumed. A periodic border trace was generated with a prescribed period, a polynomial baseline standing for anatomy, additive noise, and beat-to-beat variability, and the period was estimated from it. Recovery was scored against the prescribed period at a tolerance of 5 per cent, and *N*min for a condition was taken as the smallest number of written cycles at which at least 90 per cent of 200 trials succeeded.

The sweep covered 144 combinations, the full grid of 4 noise levels from 0.05 to 0.4 of the wave amplitude, 4 baseline strengths from zero to 2.0 times it, 3 sampling densities of 8, 16 and 32 samples per cycle, and 3 degrees of beat-to-beat variability up to 0.1 of the period. The central condition against which the headline is quoted is noise 0.1, baseline 1.0, 16 samples per cycle and variability 0.05. Reporting the whole grid rather than the central cell is what makes the value a plateau rather than a point.

3.2 How far *N*min moves with the estimator, and with the conditions

Two estimators were run over the same traces: one matched to the generating model, and one using only the fundamental of the trace. The matched estimator reaches 1.5 cycles and the fundamental estimator 2.5 cycles. The remainder of this paper uses 2.5, because the matched value assumes knowledge of the waveform that no real trace supplies, and a bound should be stated at the value a usable estimator can attain.

The figure is stable across the sweep (figure 2): 116 of the 144 cells return exactly 2.5, and the worst cell anywhere in the sweep returns 3.5.

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4. Where real protocols fall

4.1 Half of the public archive cannot be asked

Evaluating the condition needs only header fields, but they have to be published. Of 13 public collections probed in The Cancer Imaging Archive[15] for acquisition parameters, 6 publish them and 7 do not. For the latter the question cannot be put at all — not because the scans fail the condition, but because nothing in the archive says what the table was doing.

This is worth stating plainly because it bounds any future study of the same kind, including the validation of motion-correction methods that this paper is about. A collection that omits table speed, pitch, collimation and rotation time cannot support a claim about what its scans could or could not have recorded.

4.2 Table speeds and thresholds

The 192 series come from 94 patients in 5 of the 6 collections that publish the parameters, with at most 40 series taken from any one of them so that a single large collection cannot set the distribution. 4 reach that cap and the fifth contributes 32. Table speed runs from 30.0 to 158.8 mm/s, a factor of 5.3, with a median of 55.0 mm/s.

Because faster hearts write more cycles into the same distance, each protocol has a *lowest* rate it can record. Over the cardiac border alone that threshold has a median of 69 bpm, ranging from 38 to 198 bpm, and 128 of 192 series — 67 per cent — have a threshold below 100 bpm. Using the longer extent available along the descending aorta, the median threshold falls to 28 bpm and 100 per cent of series qualify.

Under the assumptions used here, most of the installed base clears the threshold (figure 3). Because the patients' own rates are not recorded, that is a statement about what those protocols could record, not a count of acquisitions that did satisfy the condition. It is enough to make section 5 worth performing.

The exceptions are instructive. A wide-detector protocol at 691.2 mm/s has a threshold of 864 bpm, and a dual-source high-pitch protocol at 1474.6 mm/s a threshold of 1843 bpm. Both are far above any physiological rate: at the assumed structure extent, and at any physiological rate, these acquisitions cross the heart so quickly that less than one full cycle is written, so no complete repetition is present in the record. They fail the cycle-count criterion of this paper by the widest margin in the survey.

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5. Meeting the threshold does not produce a period

5.1 A cohort under a frozen criterion

The cohort comprised 18 series, of which 17 yielded a volume the extraction could read; the remaining 1 is reported as a technical failure and enters no count. They are drawn from the same 192 series evaluated in section 4 — all 18 of them — one series per patient, so 18 patients contribute one each and no patient appears twice. 3 of them were the pilots on which the extraction was developed, and they are retained rather than dropped because removing them after the fact would be a choice made with the outcomes visible.

A series was admitted when the joint fit spanned at least $N\min = 2.5$ cycles, was not pinned at either edge of the physiological band, drew on at least three usable coronal levels, moved by no more than 10 bpm when any one level was dropped, and left a median residual below the amplitude it had fitted. Those thresholds were written down before any series beyond the three pilots was fitted.

One detail of that first condition matters and is easy to miss. The implementation counts cycles over the z span actually analysed, not over L, the extent of the structure that section 2 says can carry the motion. For this cohort the two are close: the analysed spans run from 256 to 654 mm with a median of 324 mm against an aortic extent of 300 mm, and the same 16 series clear $N\min$ either way. It is stated here because for the one series in section 5.4 the two differ by a factor of two.

What the fit reads is a one-dimensional trace of that border against z (figure 4). At each slice of a coronal row, the candidates are the runs above -300 HU that cross the row's midline and are at least 25 mm wide, and the border is the left edge of the chosen run. The border is followed rather than chosen independently on each slice: tracking begins in the middle of the scan where the mediastinum is widest, each slice takes the candidate nearest to the previous one, and a slice with no candidate within 6 mm is left empty rather than filled with the nearest available structure. 6 rows are attempted, at fractions 0.45 to 0.7 of the image height, and every one that yields a trace is reported rather than the clearest. A row is kept when the tracker returns a border on at least 64 slices; rows that fail that test are rows where no run above the lung threshold crosses the midline widely enough often enough, the mediastinum not being equally wide at every height in every patient. Across the cohort 3 to 6 rows survive, and the criterion below requires at least three.

The period is fitted to all levels at once. Writing $u = (z - z_0)/\lambda$, with $\lambda = ST$ the spatial period of section 2.1, and $v = (z - z_0)/\text{span}$, each level is modelled as a sinusoid on a quartic baseline, $a \cos 2\pi u + b \sin 2\pi u + c_0 + c_1 v + c_2 v^2 + c_3 v^3 + c_4 v^4$, whose coefficients are solved by least squares at each trial period; amplitude and phase are therefore free per level while the spatial period λ is common, because there is one heart and because in a helical scan the nominal acquisition time is a function of z alone, so every level shares it to the same approximation (section 2.1). The joint cost at a period is the sum over

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levels of the residual sum of squares, each weighted by the reciprocal of that level's variance, so a level with a large anatomical swing does not outvote the others. The spatial period is taken as the minimiser over 600 points spanning 40 to 120 bpm at the header's table speed; that band is physiological, and is never set by what the data prefer.

What that criterion measures is self-consistency. Leave-one-out movement and residual size both ask whether the fit holds still; neither compares the fitted rate with the patient's. No series here records one (section 5.4), so correctness was not available to be tested, and the criterion should not be read as if it had been. The count that follows is therefore a count of series meeting a self-consistency criterion. It is neither a count of correct estimates nor a count of incorrect ones, and it is not an accuracy.

3 of the 17 series met it. The header-level condition predicted 13 of them to be recoverable at a ceiling of 100 bpm, and 2 of those 13 were admitted.

The pre-specified summary was the agreement between prediction and outcome, not the admission rate. The two agreed on 5 of 17 series, 29 per cent.

That number needs reading carefully, and I state it descriptively rather than as a test. The prediction is a header-level statement about whether a rate below 100 bpm could in principle be written; the outcome is a self-consistency verdict that never sees a true rate. They are not two measurements of the same thing, so their disagreement is not evidence that either is wrong, and I attach no null distribution to it. With 17 series and two quantities that are not measurements of the same thing, the figure is not strong enough to establish a relation between criterion and acquisition parameters, nor the absence of one; I report it because it was the pre-specified summary. What the admissions are responding to is examined directly in sections 5.3 and 5.4, which is the more informative route.

5.2 How real traces differ from the simulated ones

The border traces of the real series do not merely fit worse than simulated ones, they differ in kind. Comparing 85 real traces with 85 simulated ones, 88 per cent of the real signal sits in the smooth baseline the fit removes, against 12 per cent of the simulated signal, and the spread of band power across levels is 7.5 times wider in the real traces (0.625 against 0.083). A real trace is mostly anatomy, and how much of it is anatomy varies from level to level in a way the simulation never produced.

5 properties were measured on both populations: the share of the signal removed by the quartic baseline, the fraction of the remaining power inside the physiological band, that band's enrichment over the rest of the spectrum, the rate of large slice-to-slice steps, and the drift of the wave's amplitude across the scan. Two of the five separate the populations sharply. The baseline share is the one quoted above. The other is the amplitude drift: a real trace's wave grows or shrinks by a median factor of 1.47 from one end of the scan to the other, against 1.04 in simulation, so the amplitude the fit assumes to

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be constant is not. Large steps are absent from the median trace in both populations but reach 2.0 per hundred slices at the ninetieth percentile of the real ones, against 0.0 in simulation.

These are differences, not causes. I did not put any of them back into the simulator to see whether the failure reappears, and this section should not be read as having excluded them. It establishes that the signal the estimator was designed for is not the signal it is given, and section 5.3 reports a property of the objective that is measured rather than inferred.

5.3 The objective is nearly flat, and the criterion never looked

The criterion asks whether the argmin moves. It does not ask whether the cost surface has a minimum worth finding. Those are different questions whenever the surface is flat, and the following measurement was not part of the frozen analysis; it was added after the fact, for the reason given in section 5.4.

Measure the depth of the minimum as the median cost across the whole physiological band divided by the cost at the period the fit returned. A value of 1.00 would mean the chosen period fits no better than an arbitrary one.

Across the cohort the depth is 1.072 at the median of the rejected series, ranging from 1.012 to 1.465. The 3 admitted series lie inside that range, at 1.106 to 1.142: the period chosen in an accepted series reduces the weighted residual by about a tenth relative to a typical period in the band, which is what the rejected series do as well. On this measure the admissions are not distinguishable from the rejections.

They are also not consensus (figure 5). In each of the 3 admitted series the usable coronal levels — 5, 5 and 6 of them — fitted independently, prefer rates whose spread, the largest minus the smallest within that series, is 44 to 59 bpm. The joint fit imposes one period on levels that individually disagree by half the physiological range, and returns their weighted compromise.

What follows from this is about the criterion, and I am careful about how far it goes. Two things are measured: the minima the criterion accepted are no deeper than those it rejected, and the levels it combined did not agree. Neither measurement uses a true rate, so neither can show that the three admitted rates are *wrong*. What they do show is that the criterion admitted them without evidence that they are right, which is a different and weaker claim than the one a reader would take from the word "recovered".

A natural explanation of how that happens is that a flat objective leaves the minimiser unpinned, so removing one level moves it very little, and the leave-one-out test then reports stability that reflects the shape of the objective rather than the strength of a signal. That account is consistent with everything reported here — shallow minima, disagreeing levels, and stable leave-one-out values occurring together — but co-occurrence is not demonstration, and I have not constructed the

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experiment that would separate the flatness from the stability. I therefore offer it as an explanation to be tested, and rest the conclusion on the measurements rather than on the mechanism.

One reading it does weaken is that the extraction simply locks onto the wrong peak. With the best period fitting only about a tenth better than a typical one, there is no peak in these objectives sharp enough for that description to be apt.

5.4 A candidate session with a recorded rate, which could not be used

A criterion that cannot be compared with a true rate can be characterised but not scored for accuracy, so I searched for a session that records one. Of 156 collections in The Cancer Imaging Archive, 102 hold CT and 57 studies are described as a cardiac examination; 2 sessions record a rate in the header, and 1 contains both a recorded rate and a free-running helical series of the chest.

The rate is not where a reader would look for it. `HeartRate (0018,1088)` is empty throughout; the scanner writes the rate into the free text of `ScanOptions (0018,0022)` as a minimum, maximum and average in bpm. A survey restricted to the standard cardiac fields concludes that no public CT records the rate, and is wrong; the count in section 4 was repeated over 11 routes for that reason. In that session a gated acquisition records 75.0 bpm, between 72.0 and 78.0 over its own duration, and a helical series covering 630 mm at 208.7 mm/s begins ten seconds later.

The series could not be used. The extraction did not return the anatomical border the method is defined on. Its tracked position lies at 0.9 per cent of the image width, where a mediastinum in this cohort lies between 41 and 48 per cent, and at one coronal level the value is constant to numerical precision across the whole scan. What structure it locked onto instead is not established. Nothing in the pipeline indicated the failure: the fit converged, the residual test passed and the leave-one-out values clustered, and only two levels departing to the edge of the physiological band kept the series out. The run is reported in the supplementary material with the protocol that was fixed before it, and it does not enter any estimate of recovery accuracy.

Having found it in one series, I measured it in all of them. Of the 17 analysed cohort series, 2 sit within 10 per cent of the image edge and 2 more sit closer to the edge than to the middle. All of the affected series were rejected by the frozen criterion, and the 3 admitted series lie at 41 to 43 per cent. The measurement is made per coronal level and not per series, because a series median near the middle could hide a single level out at the edge, which is what happened on the reference case. All 16 usable levels of the three admitted series lie between 35 and 48 per cent of the image width, and figure 4 draws the border on the image for the one of the three whose border sits deepest. What that establishes is the absence of the specific failure it was written to detect, a border out at the surface. It is not a test of anatomical correctness: a position near the middle of the image is where the mediastinum is, and also where several other structures are. Section 5.5 examines all three admitted

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series as images and reports what this check misses. The check that measures this, `analysis/check_border_position.py`, did not exist while the cohort was being analysed.

5.5 What the admitted series were actually measuring

To find out what the three admitted series were measuring, every usable coronal level of each — 16 in all — was drawn as a reformat with its tracked border on it and examined. The images are in the supplementary material, and `analysis/audit_admitted_levels.py` produces them and the numbers below. Neither alters the frozen outcome of section 5.1.

No admitted level is on the skin. None of the 16 lies within 10 per cent of the image edge, so the failure of section 5.4 does not recur here. Looking at the reformats shows a different failure that the position check cannot see. Beyond the lung base, and past the apex, the tracker still returns a border although there is no lung there: both sides of it are soft tissue at about 0 HU, and no lung–mediastinum interface exists. Classifying every returned point by the attenuation on each side — aerated lung one side, soft tissue the other — 2610 of 2997 points, or 87 per cent, sit at such an interface. No level is wholly clean, and on the worst 66 per cent are.

The consequence is not evenly spread, and for one series it is serious. The points that do sit at a lung interface span a shorter stretch of z than the levels as a whole, and the criterion of section 5.1 counted cycles over the whole. Re-counted over the valid stretch at the period each fit returned, 1 of the 3 admitted series falls below $N\mathrm{min}$: its valid stretch is 61 per cent of the span analysed, and the cycles written across it are 1.68 rather than the 2.77 the criterion recorded. That series met the cycle-count condition on paper and does not meet it over the part of the scan where the extraction is on the structure the method is defined on.

Two further properties are visible in the reformats and were not measured before: the traces are sparse, with 88 per cent of slices returning no border at all on the worst level, and the tracked line is smooth where the method expects an undulation — the same thing section 5.3 reads off the objective. For all three admissions, then, the fitted period rests on a partial trace of a partly correct border.

6. Discussion

This study establishes four things, and they rest on evidence of different kinds. The two sampling requirements follow from the geometry of a helical acquisition, and their product is fixed by the structure extent and the reconstruction interval, so table speed redistributes a budget it cannot enlarge. The number of cycles one estimator needs was measured rather than assumed. Applied to 192 real series through their headers, and under stated assumptions about extent and rate, the requirements are satisfiable for most of them. And on images, where no rate is recorded, a criterion that asks only whether a fit holds still admits series whose objective is nearly flat, so such a criterion does not

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establish that a period was found — the most transferable of the four, and the one this section returns to last.

Two of those results share a structure that is worth stating once.

In section 5 a stability criterion admitted 3 series whose coronal levels, taken separately, preferred rates spread over 44 to 59 bpm, on cost surfaces whose minima are no deeper than those of the series it rejected. In the supplementary material an estimator trained only inside the recoverable regime reported a standard deviation of 0.045 while achieving 0 per cent accuracy outside it. In both, a diagnostic internal to the method reported confidence in a regime the method could not serve, and in neither did the method's own output indicate a problem.

The difference between them is instructive. The estimator's failure is a training failure and has a training remedy: shown the boundary during training, the same architecture widens its reported uncertainty to 0.239 and reports it where an answer is not available, while reaching 85 per cent accuracy at two cycles — the calibration is what changes, not a rule about when to answer. The criterion's failure is a design failure, and what section 5.3 measures against it is diagnostic rather than decisive: a stability statistic on its own does not say whether the objective it was computed on has a minimum worth being stable about. Reporting the depth of the minimum, relative to the median over the searched band, alongside the disagreement between levels would have made the ambiguity in the three admissions visible where the stability statistic concealed it. It would not have decided them. No acceptance threshold on depth is proposed here and none is validated: doing so would need cases whose rates are known, which is what this study lacks. The depth is something to report alongside a fit, not a test the fit can pass.

None of this weakens the geometry. What follows separates the three positive claims by the strength of the evidence behind each, because they should not be read as one.

The budget identity $N\,n = L/dz$ follows from the definitions and holds for any helical acquisition. It is the firmest thing here.

$N\mathrm{min} = 2.5$ cycles is an empirical threshold, measured for one estimator reading one family of simulated traces at one tolerance and one success rate. It is not a universal limit: the matched estimator of section 3.2 reaches 1.5 cycles on the same traces, and the network of the supplementary material (S4.2), trained on that family, is accurate at two. What the bound supports is a practical screen — under the conditions studied here, an acquisition writing appreciably fewer cycles than this should not be expected to yield a period — rather than a statement about every possible estimator.

The header-level evaluation of section 4 is narrower still than it first appears. Of 192 real chest CT series, 67 per cent have a threshold below 100 bpm at an assumed cardiac extent of 120 mm. Because

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the actual rates are not recorded, that is a statement about what those protocols could record for a patient beating fast enough, not a count of acquisitions that did satisfy the condition.

What sections 5.3 to 5.5 add is that clearing the threshold does not by itself produce a period: the objective is nearly flat, the stability the criterion measures is not evidence that a period was found, and on inspection one of the three admissions does not clear the cycle-count condition over the stretch where its extraction is on the intended structure. The bound is a screen, not a promise.

Where this would sit in a route to clinical use. A component of the undulation attributable to cardiac motion would, if it could be separated, carry phase as well as period, its maxima and minima falling on opposite parts of the cycle; that is the hypothesis which makes reading timing out of an existing scan of interest, and it is not tested here. What the present study contributes to that route is narrow and comes early in it. The sampling requirements say which acquisition conditions meet the criterion defined in this paper, and are computable from the header before any image is opened, so they can be used to select the acquisitions on which a future estimator is developed or tested. The image evaluation says what an estimate from such a scan does and does not license: a self-consistent fit is not a validated one, and the depth of the objective and the agreement between levels are the quantities to report alongside it. Everything beyond that needs its own validation and receives none here. Identifying which extremum is end-diastole and which end-systole requires a phase reference this study does not have; phase-resolved reformatting requires that the displacement survive reconstruction well enough to sort slices by, which section 2.1 notes is untested; and volumetry or an ejection fraction requires all of that plus a comparison against an accepted measurement. None of those steps is supported or excluded by the results reported here.

Two consequences follow for anyone who would validate such a method. These data did not permit reference validation: one session in 156 collections of The Cancer Imaging Archive holds both a recorded rate and a free-running helical series, and 7 of 13 collections probed do not publish acquisition parameters at all, so for those the requirements cannot even be evaluated. And the failure mode of the supplementary material — a confident report from outside the regime a model was trained in — exists and is invisible from the model's own output, which is an argument for measuring it in a deployed estimator rather than a finding about any particular one.

7. Limitations

The bound was measured against prescribed periods, and applied where no reference period is recorded. $N\mathrm{min} = 2.5$ cycles was obtained by simulation over 144 parameter combinations, in which the period is prescribed and recovery can be scored against it. It was then evaluated against 192 real series through their acquisition parameters alone, and applied to 17 image series in which no reference period is recorded. Exactly one real reconstructed image in the archive can be scored against a known

period, and on that series the extraction did not return the intended border, so it could not be used. No real reconstructed image in this study is scored against a known period.

A fourth candidate cause was found during this revision. The cohort outcome remains consistent with the acquisitions writing too little of the cycle, with the fit being inadequate on real anatomy, or with cardiac motion departing too far from a fixed period, and this study does not separate those three. To them must be added the extraction not being on the intended structure, which sections 5.4 and 5.5 show for the rejected series and for part of every admitted level respectively. Nothing in the pipeline indicated either. Before this revision the study had no check that it was measuring the right thing.

What the evidence does not support is any claim that no method could recover the period. I can say that this extraction finds no peak worth locking onto in the traces where it is on the mediastinum, and that where it is not on the mediastinum the question was never asked. The result is a property of one border tracker and one joint sinusoidal fit on seventeen series. It does not bear on other ways of reading the same undulation — selecting extremal levels directly, correlating whole reformats, or estimating phase without first estimating a period — nor on any subsequent step such as phase-resolved reconstruction or volumetry, none of which is attempted here.

No reference heart rate exists in the cohort itself. None of the 18 series retrieved for image analysis records the rate by any of the 11 routes I checked: the 10 standard DICOM cardiac fields, and the free text of ScanOptions, where a Siemens cardiac acquisition writes the rate while leaving the standard fields empty. The one session that does record it is not in the cohort, and its rate is read from a gated series acquired ten seconds before the series under test rather than from that series itself; the recorded minimum and maximum, 72.0 and 78.0 bpm, are the scanner's own measure of how far the rate moved during an acquisition of that length, and are carried into the tolerance rather than ignored. The archive search that found it keys on study descriptions naming a cardiac examination, so 1 is a lower bound.

What would settle it, and what would not. Scans of a moving phantom driven at known periods, repeated across a range of table speeds and pitches, would score recovery against truth in real reconstructed images and would separate the acquisition and extraction causes from each other. I did not have access to a scanner and a motion phantom, and this work does not include such a study. A digital phantom carried through helical forward projection and reconstruction would reach part of the same ground, testing the extraction on images rather than on traces and allowing acquisition conditions to be swept; it would not settle the third cause, because the beat-to-beat variability of such a phantom is an assumption supplied by the modeller rather than a property of a patient.

Three analyses here are post-hoc, and their status differs from the rest. The depth measurement of section 5.3, the reference case of section 5.4 and the anatomical audit of section 5.5 were all added after the cohort had been frozen, fitted and written up, in response to editorial feedback on a

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presubmission enquiry. The cohort outcome reported in section 5.1 is the frozen one; the audit re-counts cycles for interpretation and does not alter which series were admitted. For the reference case the acceptance interval was fixed in the repository before the images were downloaded, and the repository history shows that order; I make no claim to an independently attested timestamp, because a local file and a local commit are both under the author's control.

8. Conclusions

The axial coordinate of a helical CT is a time axis whose scale the header states, so a periodically moving structure writes its period into the volume as a spatial period. Recovering it needs enough cycles written and enough samples within each, and those two requirements multiply to L/dz: a protocol cannot buy more cardiac timing by changing its speed, only spend a fixed budget differently. That identity follows from the definitions. The constant the window depends on was measured rather than assumed, and it is empirical and estimator-dependent — 2.5 cycles for an estimator using only the fundamental of the trace, 1.5 for one matched to the waveform — so it serves as a practical screen under stated assumptions about the structure extent and the rate, not as a limit on every estimator. Evaluated that way, 67 per cent of 192 real chest CT series could record a rate below 100 bpm at an assumed cardiac extent, and 100 per cent over the aortic extent; because the patients' rates are not recorded, that says what those protocols could do rather than what they did.

Meeting the criterion does not produce a period. Under a criterion frozen in advance, 3 of 17 image series were admitted, and the admissions are not evidence of recovery: their cost minima are no deeper than the rejections', at 1.106 to 1.142 against a rejected median of 1.072, and their coronal levels individually prefer rates differing within a series by 44 to 59 bpm. An anatomical audit of all 16 of their coronal levels adds that 87 per cent of the tracked points sit at a lung interface, and that one of the three, re-counted over the stretch where its extraction is valid, writes 1.68 cycles rather than the 2.77 the criterion recorded. Neither can they be shown to be wrong, because no rate is recorded: the one archive session that records one could not be used, since there the extraction did not return the intended anatomical border and nothing in the pipeline said so. The image evaluation is therefore exploratory, and what it establishes is that a self-consistency criterion does not measure whether a period was found.

Two practical consequences follow. The bound is worth computing, because it comes free from the header, but failing it means only that the recovery rate defining $N\mathrm{min}$ — 90 per cent of trials within 5 per cent — was not reached under the conditions studied, and not that the acquisition holds no signal or that no estimator could succeed. And a fitted period should be reported together with the depth of the minimum it sits in and with a check that the structure fitted was the intended one. This study began with neither, and adding them is what changed how it reads its own results. The auxiliary

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simulation of the supplementary material makes the same point about an estimator's declared uncertainty: a model trained only where recovery was possible reported its smallest uncertainty exactly where it was never right, so that uncertainty too is a quantity to be checked rather than taken at face value. Taken together, the sampling requirements and the image evaluation are useful for the next step rather than being a conclusion about it: they say which acquisitions a future period or phase estimator could be tested on, and what its output would have to show before being believed. Whether a period can be recovered accurately from reconstructed non-gated images is not settled here, and the uses that would follow from it — phase identification, phase-resolved reconstruction, ventricular volumetry — are neither demonstrated nor ruled out.

Until that is settled, a reader who would use such an estimate rather than build one has two things to ask for before acting on it, and both are available now. The first is whether the acquisition could have carried the timing at all: the header answers that before any image is opened, and it costs nothing. The second is whether the reported period sits in a deep minimum of the structure it was meant to follow, which the fit and one reformat answer together. The series admitted here met the first on paper and did not meet the second — no level tracked the intended border throughout, and re-counting over the stretch where it did put one of the three below Nmin — while the criterion that admitted them looked at neither. A cardiac period or phase offered from a routine non-gated chest CT should therefore not be carried into clinical use on the strength of a stable fit alone.

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Acknowledgements

The research question, the physical argument of section 2, the study design and the decision rules recorded in the repository are the author's and predate the assistance described here. Generative AI (Claude, Anthropic) was used as a tool in preparing this work: to write and test analysis code, to carry out the survey of the public archive reported in section 5.4, to produce the figures from the frozen result files, and to draft manuscript text from results that already existed. No AI system contributed a research idea, chose a decision rule, or determined a result. Every number in this paper is resolved at build time from a result file produced by code in the public repository, every reference is resolved from its DOI rather than written out, and the author has checked the manuscript against those files and takes full responsibility for its content, including the parts drafted with assistance.

Statements and Declarations

Ethical considerations

This study used only publicly available, de-identified imaging distributed by The Cancer Imaging Archive under Creative Commons licences, and generated no new patient data. Ethical approval was

therefore not required and none was sought: secondary analysis of de-identified public data does not constitute human subjects research. Approval for the original acquisitions rests with the contributing institutions and is recorded by the archive. Patient identifiers and series instance identifiers appearing in the released results are the archive's own de-identified ones.

Consent to participate

Not applicable. No participant was recruited or contacted for this study, which analysed images already published in de-identified form.

Consent for publication

Not applicable. The article contains no identifiable individual data.

Declaration of conflicting interest

The author declares the following potential conflict of interest with respect to the research, authorship and publication of this article: the author is the representative of LISIT Co., Ltd., a provider of medical imaging analysis services, and this work was carried out within that company. The company received no funding for the study, played no part in its design, analysis or reporting, and markets no product based on the method examined here. The author declares no competing interest beyond that affiliation.

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Funding statement

This work received no external funding. It was carried out within LISIT Co., Ltd., using the author's own company's equipment as support in kind.

Data availability

All analysis code, the frozen decision rules and every result file from which the figures and numbers in this paper are computed are available at https://github.com/Institute-of-One/non-ecg-core, at the tagged release v0.1.1, which accompanies this manuscript, and archived at Zenodo, https://doi.org/10.5281/zenodo.23051189; the concept identifier 10.5281/zenodo.23034327 resolves to the most recent version. A clean copy of that release, unpacked into an empty directory and installed from `requirements-core.txt` alone, rebuilds every number in this paper to the same value and draws every figure that does not show an image; `verify_release.py`, in the release, is the script that checks it. Five figures do show one — figure 4 in the main text and figures S1 and S4 to S6 in the supplementary material, all of them coronal reformats — and those need the archive series themselves. No imaging data is redistributed here: the code retrieves it from The Cancer Imaging Archive by series identifier, which requires the full `requirements.txt` and a network connection. The collections from which series were taken are cited as data: Anti-PD-1 Immunotherapy Lung [16], Stony Brook University COVID-19 Positive Cases [17], CPTAC Lung Adenocarcinoma [18], A Large-Scale CT

and PET/CT Dataset for Lung Cancer Diagnosis [19], the RIDER coffee-break lung CT collection [20] and VAREPOP-APOLLO [21].

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Figure legends

Figure 1. Cycles written across the heart against samples per cardiac cycle, at 60 bpm, on logarithmic axes. Each protocol is a point; the grey lines are the budget $N\,n = L/dz$ for three reconstruction intervals, and a protocol can move only along its own line. The shaded corner is the region in which both conditions hold, bounded below by $N\mathrm{min} = 2.5$ cycles and on the left by $n = 8$ samples per cycle. Blue points satisfy both conditions, red points fail at least one. The two 2003 four-row protocols are the acquisitions from which this problem originally arose; the modern chest protocols sit an order of magnitude to the right and below the bound, having spent the whole budget on resolution within a cycle.

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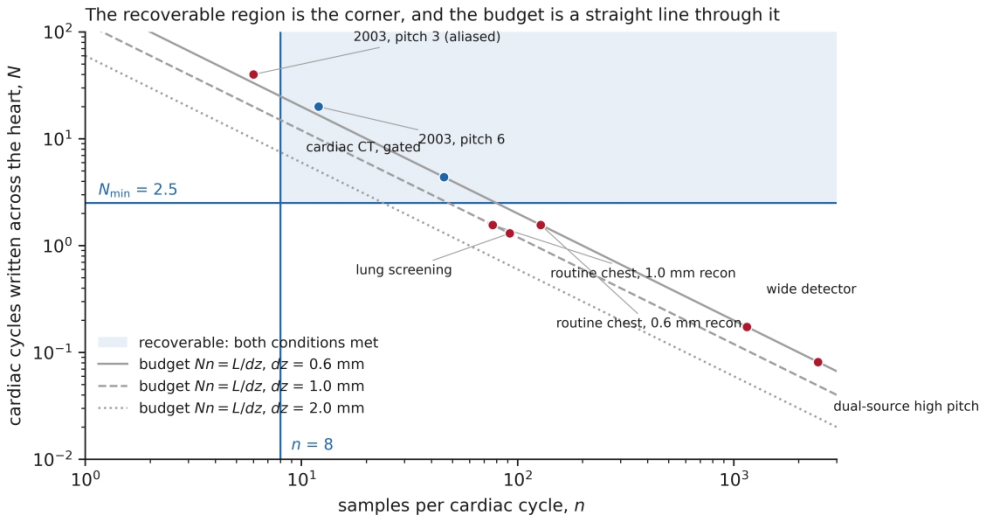
Figure 2. Left: the fraction of 200 trials in which the prescribed period was recovered to within 5 per cent, against the number of cycles written, for the matched and the fundamental estimator at the central parameter condition. *N*min is where each curve crosses 90 per cent, marked by the dashed verticals at 1.5 and 2.5 cycles. Right: *N*min obtained independently in each of the 144 cells of the sensitivity sweep. The constant is a plateau, not a point estimate. The regime that degrades it is coarse sampling within a cycle combined with high noise, which is the corner in which the budget identity of section 2.3 says the two requirements are already competing for the same fixed product.

Figure 3. The lowest heart rate each of the 192 real chest CT series could record, computed from its header alone at *N*min = 2.5 cycles, over the cardiac border (L = 120 mm) and over the descending aorta (L = 300 mm). The vertical line is 100 bpm: series to its left could record any physiological rate above their own threshold.

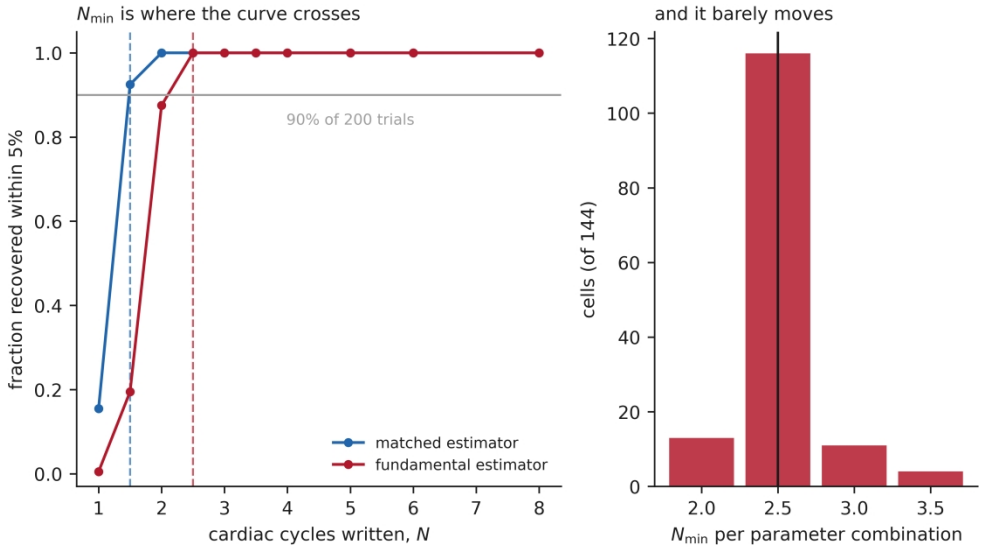
Figure 4. A coronal reformat from one of the admitted series, at its own aspect — one millimetre along z is one millimetre along x — with the tracked border drawn on it. The border follows the mediastinum between the two lungs, which is the structure the method is defined on. Its position, 43 per cent of the image width here, is what section 5.4 measures for every series.

Figure 5. Left: depth of the minimum, defined as the median cost across the physiological band divided by the cost at the period returned, for the series the frozen criterion rejected and for those it admitted. A value of 1.0, the dashed line, would mean the chosen period fits no better than an arbitrary one. The admitted series lie inside the range of the rejected ones. Right: for each admitted series, the rate each usable coronal level prefers when fitted alone (grey), against the rate the joint fit returns (cross). The joint fit reports a compromise among levels that do not agree.

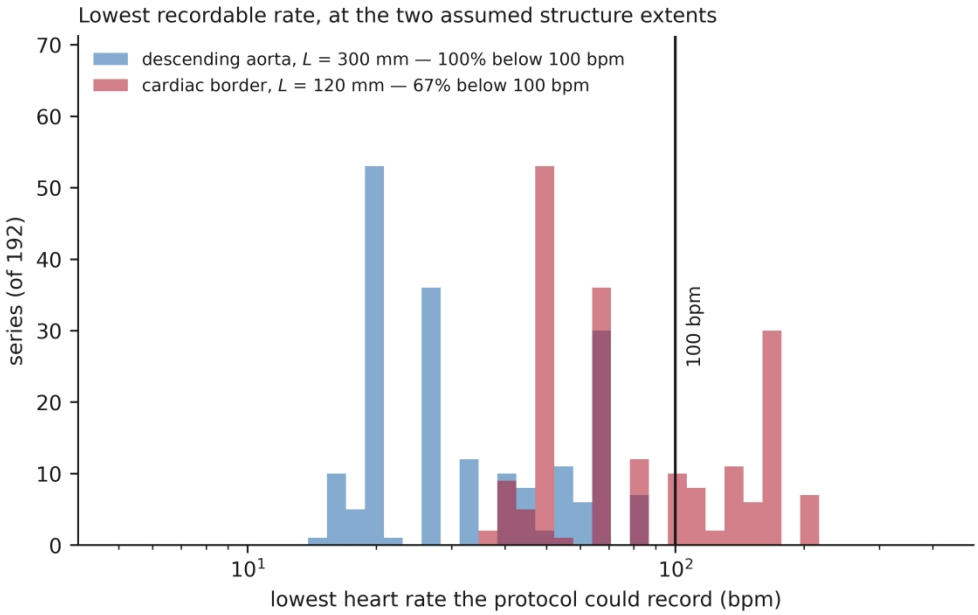
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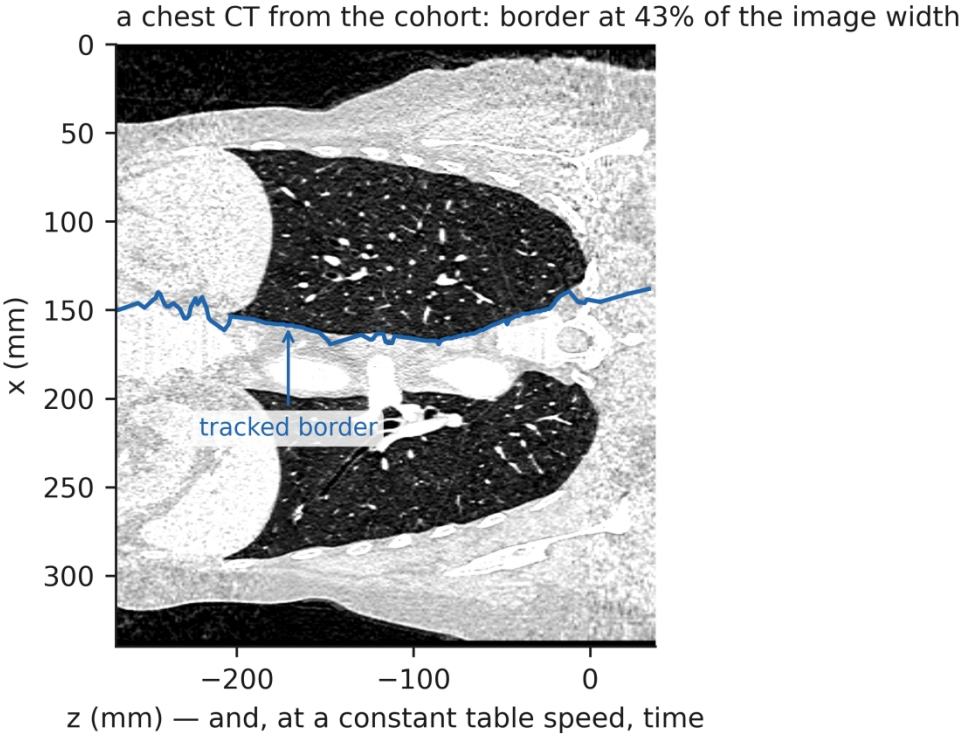
466x275mm (236 x 236 DPI)



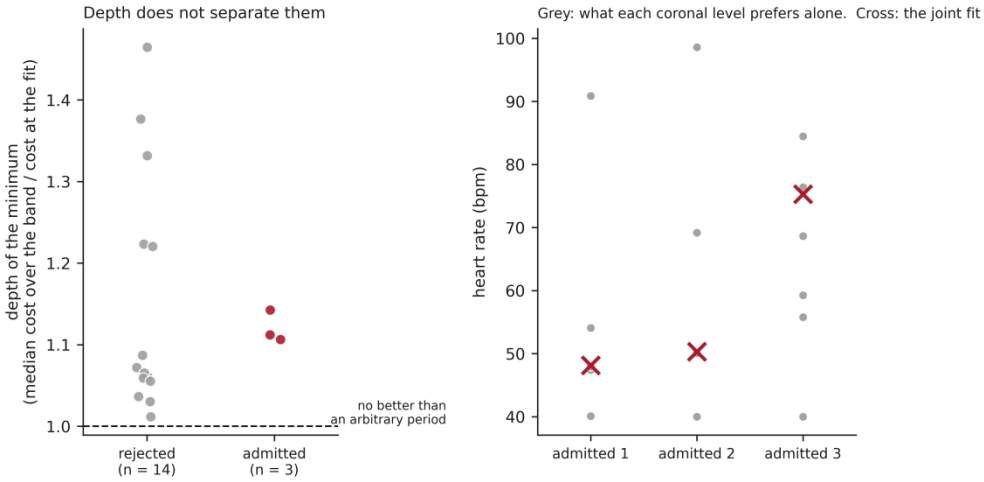
447x279mm (236 x 236 DPI)



391x274mm (236 x 236 DPI)



295x253mm (236 x 236 DPI)



511x279mm (236 x 236 DPI)

Supplementary material

Cardiac period estimation from non-gated helical CT: sampling criteria and exploratory evaluation

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This material records a run that is referred to in section 5.4 of the main text and is not used in any estimate there. It is included because the protocol was fixed before the data were fetched and the result should be visible whatever it turned out to be, and because the way it failed is the reason the main text now reports where the tracked border lies.

S1 The session, and the protocol fixed before it

A survey of 156 collections of The Cancer Imaging Archive found 1 session holding both a recorded heart rate and a free-running helical series of the chest: collection VAREPOP-APOLLO, patient AP-26JK, study described as a cardiac CTA. A gated acquisition in that session records 75.0 bpm, with a minimum of 72.0 and a maximum of 78.0 over its own duration, written into the free text of `ScanOptions (0018,0022)` rather than into `HeartRate (0018,1088)`, which is empty. A helical series covering 630 mm at 208.7 mm/s, lasting 3.02 s, begins ten seconds later.

Before the images were downloaded, an interval for agreement was fixed at five per cent of the recorded rate, 71.25 to 78.75 bpm, in `analysis/tcia_search/validation_protocol.md`. Five per cent is the tolerance under which *N*min was measured; the range the scanner itself recorded is narrower, at four per cent of the mean. The unmodified pipeline was then run.

S2 What the run returned, and why it is not used

The pipeline returned 67.4 bpm, which is -10.2 per cent from the recorded rate and outside the interval. The frozen criterion rejected the series, on a leave-one-out spread of 28.1 bpm.

Neither figure is evidence about cardiac timing, because the extraction did not return the intended anatomical border (figure S1). The tracked position lies at 0.9 per cent of the image width, where a mediastinum in the cohort lies between 41 and 48 per cent, and at one coronal level it is constant to numerical precision across the whole scan. The acquisition runs from the chin to the pelvis with the arms at the sides; the method takes the left edge of the soft-tissue run crossing the midline of a coronal row, which is the mediastinal border only where a lung breaks that run.

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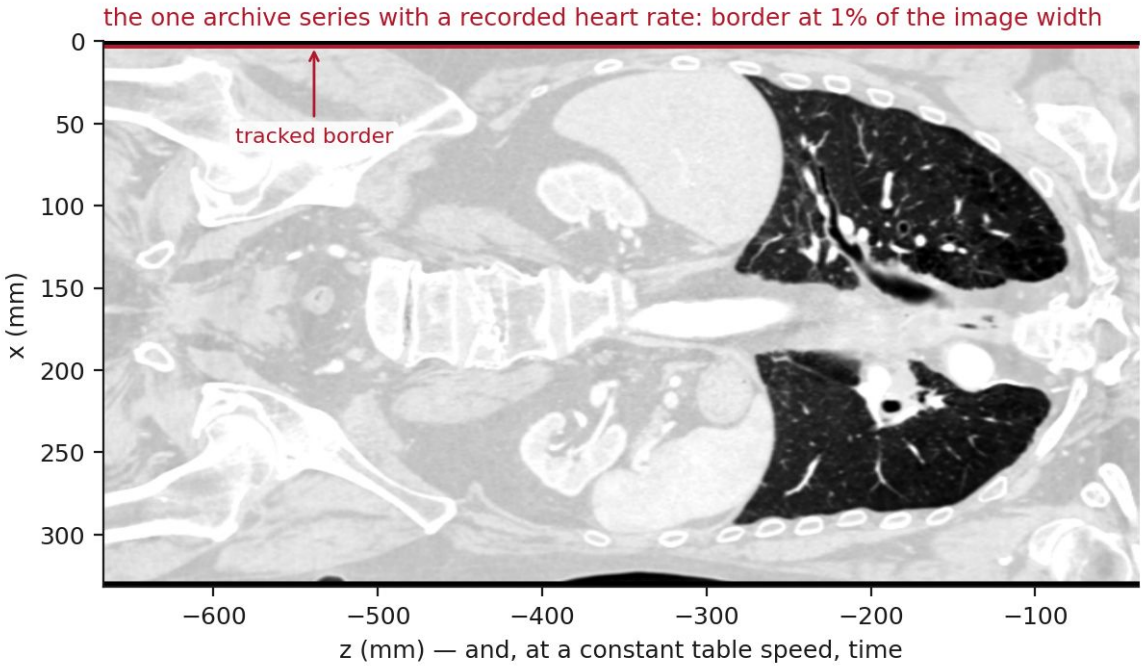


Figure S1. The coronal reformat of the series under test, at its own aspect, with the tracked border drawn on it. Compare figure 4 of the main text, where the same code follows the mediastinum between the two lungs.

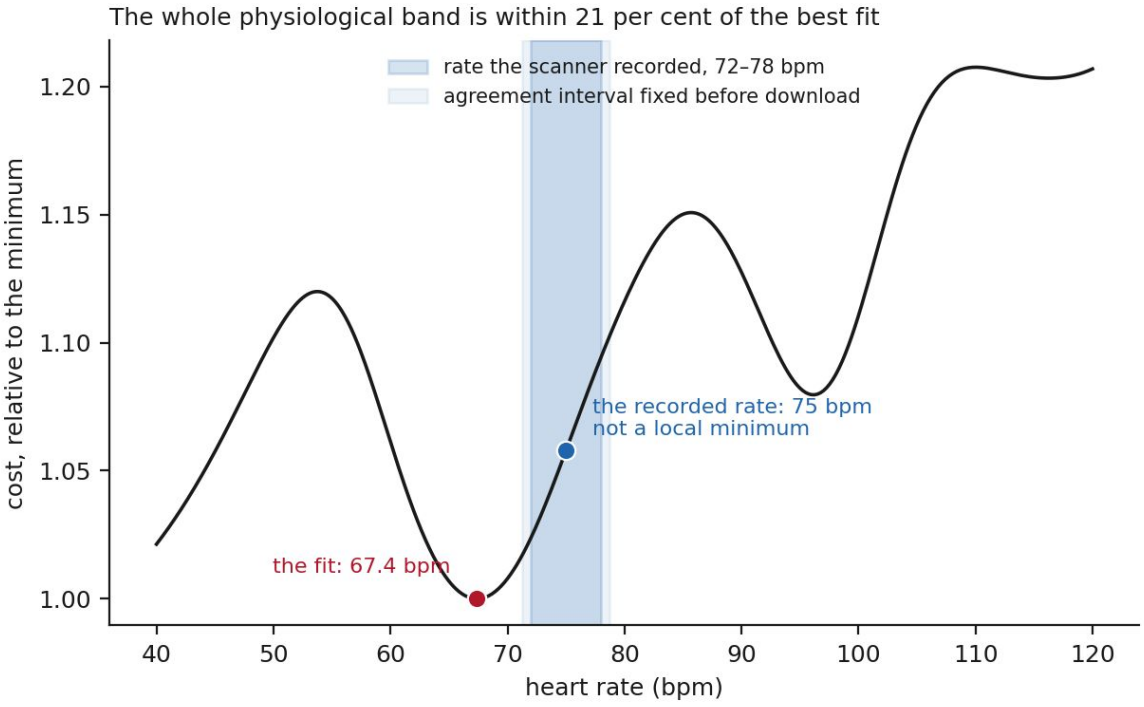


Figure S2. The objective of the joint fit across the physiological band for that series, normalised to its minimum, with the recorded rate and the interval fixed in advance marked. The recorded rate is not a local minimum, and the whole band lies within a fifth of the best fit. Since the trace is not of the intended structure, this says nothing about whether a cardiac period could have been recovered from a scan of this geometry; it is shown because the objective looked no different from those of section 5.3 until the image was drawn.

S3 What followed

`analysis/check_border_position.py`, which measures where the tracked border lies in every cached series, was written after this run and did not exist while the cohort was analysed. Applied to the cohort it finds 2 of 17 series within 10 per cent of the image edge, and 2 more closer to the edge than to the middle. All were rejected by the frozen criterion, and the 3 admitted series lie at 41 to 43 per cent.

S4 The uncertainty an estimator declares outside its training range

Sections 2 to 5 of the main text concern what an acquisition contains. This section asks a smaller and separate question, in simulation throughout: when an estimator is asked for an answer outside the range it was trained on, what does it report about its own reliability? It is included because an estimate is acted on together with whatever uncertainty accompanies it, so that uncertainty is itself something that has to be checked.

S4.1 The experiment

An estimator was trained on the same simulated traces to predict the period and, alongside it, its own standard deviation.[1] Modern networks are systematically overconfident,[2] and more so away from the training distribution;[3] what is measured here is one such shift, produced by moving the test points outside the range of the target rather than by changing the dataset. Each trace has the same quartic baseline removed as elsewhere in this paper, is divided by its own standard deviation and is resampled to a fixed length, and the target is the period as a fraction of the trace length — the reciprocal of the cycles written — so the standard deviations quoted below are in those units. The model is a small multilayer perceptron with separate heads for the mean and the log variance, trained on 60000 traces by minimising the Gaussian negative log-likelihood; the architecture, the optimiser and the training schedule are set out in the supplementary material and in `analysis/learned_estimator.py`. Accuracy is scored on 2000 independent traces at each of nine cycle counts within the same 5 per cent tolerance as section 3, and calibration is the ratio of the error actually made to the standard deviation reported, so that 1 is honest and larger is overconfident.

Two models were trained, differing only in the range of traces they saw: one on 2.5 to 8.0 cycles, at or above the bound, and one on 0.5 to 8.0 cycles.

Because the target is a reciprocal, testing below the bound is testing outside the first model's training range. It saw targets between 0.125 and 0.4; at two cycles the correct answer is 0.5 and at half a cycle 2.0, five times the largest value it was ever shown. Its failures there are failures of extrapolation and establish nothing about the information in the trace. Both models are trained and tested on simulated traces only, and nothing in this section is evidence about real chest CT.

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S4.2 What the two models report

At and above the bound the model trained there works, reaching 100 per cent accuracy. Below the bound it is wrong in every trial — 0 per cent accuracy at every cycle count tested — while reporting a standard deviation no larger than 0.045, so its calibration ratio there runs from 12.65 to 42.0: the error it makes is between twelve and forty times the uncertainty it declares. The declared uncertainty tightens as the accuracy stays at zero (figure 6). As the number of written cycles rises towards the bound the reported standard deviation falls and the accuracy does not move, so at two cycles the model is at its most certain and still never right, and nothing in what it returns marks those points as different from the ones it can serve.

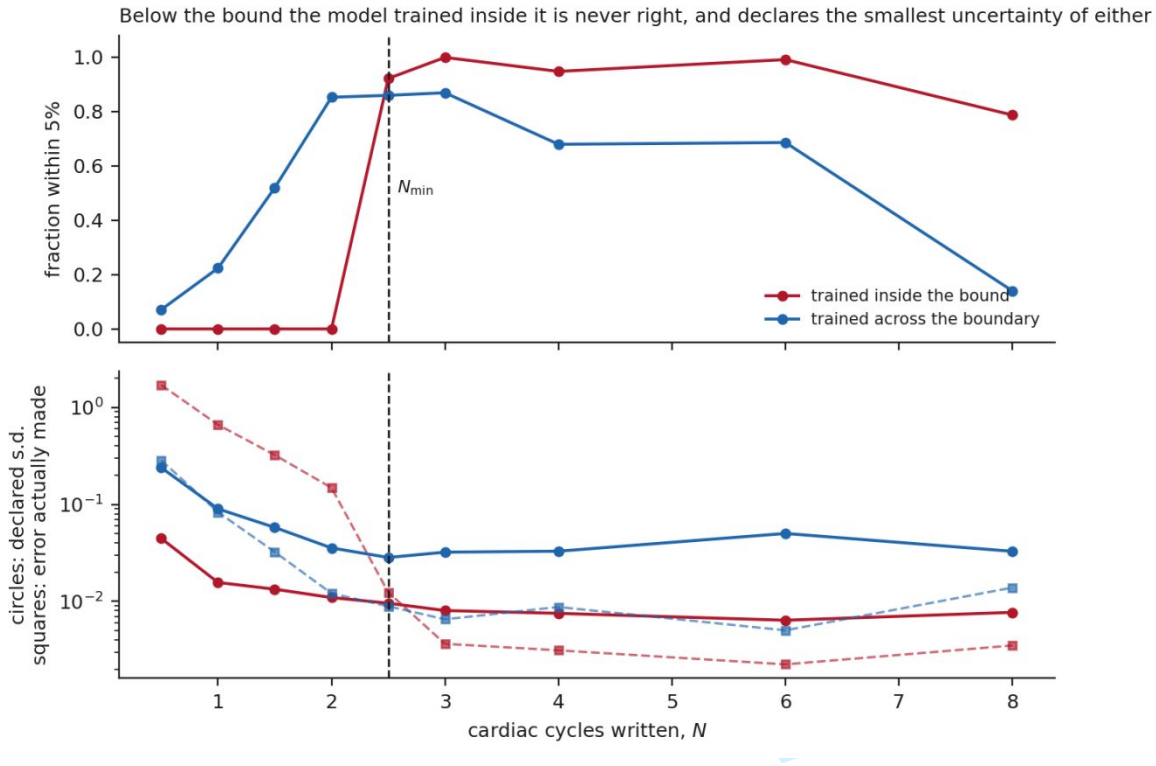


Figure S3. Top: the fraction of estimates within 5 per cent of the prescribed period, against cycles written, for a model trained only at or above the bound and for one trained across it. Bottom, on a logarithmic scale: the standard deviation each model declares (circles) and the error it actually makes (squares). Below *N*min, marked by the dashed vertical, the model trained inside the bound declares the smallest uncertainty of either while its error is largest — the gap between its circles and its squares is the overconfidence. Those test points lie outside the range of the target that model was trained on. The model trained across the boundary keeps the two together.

The model trained across the boundary behaves differently below the bound: its reported standard deviation widens to 0.239 as the cycles run out and its calibration ratio falls into 0.34 to 1.4, which is honest to conservative. Neither model abstains — both always return an estimate, and what differs is the uncertainty attached to it — so whether a scan is declined or flagged remains a decision for whoever deploys such an estimator, and one available to them only if the uncertainty means something.

Two qualifications limit what this comparison supports. The across-trained model also becomes *accurate* below 2.5 cycles, reaching 85 per cent at two. That is not a violation of the bound but the bound at a different value: *N*min depends on what the estimator knows about the waveform, and a network trained on this exact family of traces has been handed knowledge it would not have in a patient, which is why 2.5 remains the value at which I state the bound. Second, above the bound that model is the less accurate of the two, 87 against 100 per cent. That pair of numbers describes one architecture under one training protocol on one family of traces. It is a reason to measure the trade-off in a given estimator, not a general result that calibration costs accuracy.

S4.3 The estimator in full

Each trace has the same quartic baseline removed as every other estimator in this paper, is divided by its own standard deviation, and is resampled to 128 points, so the model sees one input shape and no absolute scale. The target is the period as a fraction of the trace length, that is the reciprocal of the cycles written; it is dimensionless, and the standard deviations reported in section 6 are in those units.

The model is a multilayer perceptron with three hidden layers of 256 units and rectified linear activations, ending in two linear heads for the mean and the log variance. It is trained by minimising the Gaussian negative log-likelihood with Adam for 40 epochs at a batch size of 256 on 60000 simulated traces. Accuracy is scored on 2000 independent traces at each of nine cycle counts, within the 5 per cent tolerance used in section 3. The learning rate and every remaining setting are in `analysis/learned_estimator.py`.

Two models were trained under this identical recipe, differing only in the range of cycle counts their traces spanned: 2.5 to 8.0 cycles for the one trained at or above the bound, and 0.5 to 8.0 for the one trained across it.

S5 The admitted series, level by level

Every usable coronal level of each series the frozen criterion admitted, drawn as a reformat at its own aspect with the tracked border on it. These are the images behind section 5.5. Each panel gives the level's position across the image, the excursion of its border, the share of slices for which the tracker returned nothing, and, where it occurs, the longest run of identical values.

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admitted series 1: 8411021488020526

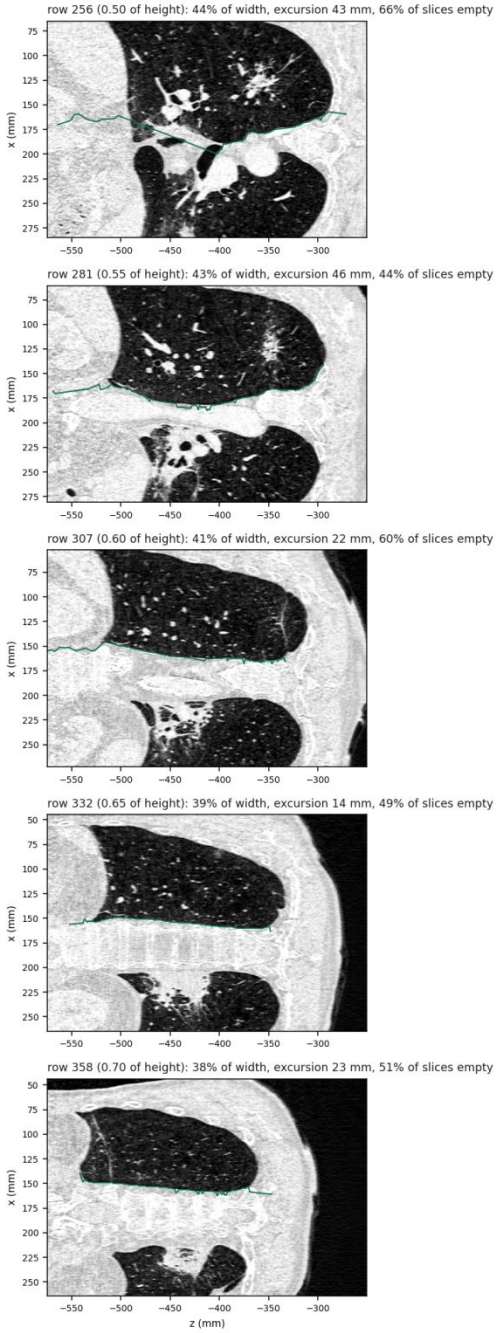


Figure S4. Admitted series 1, five usable levels.

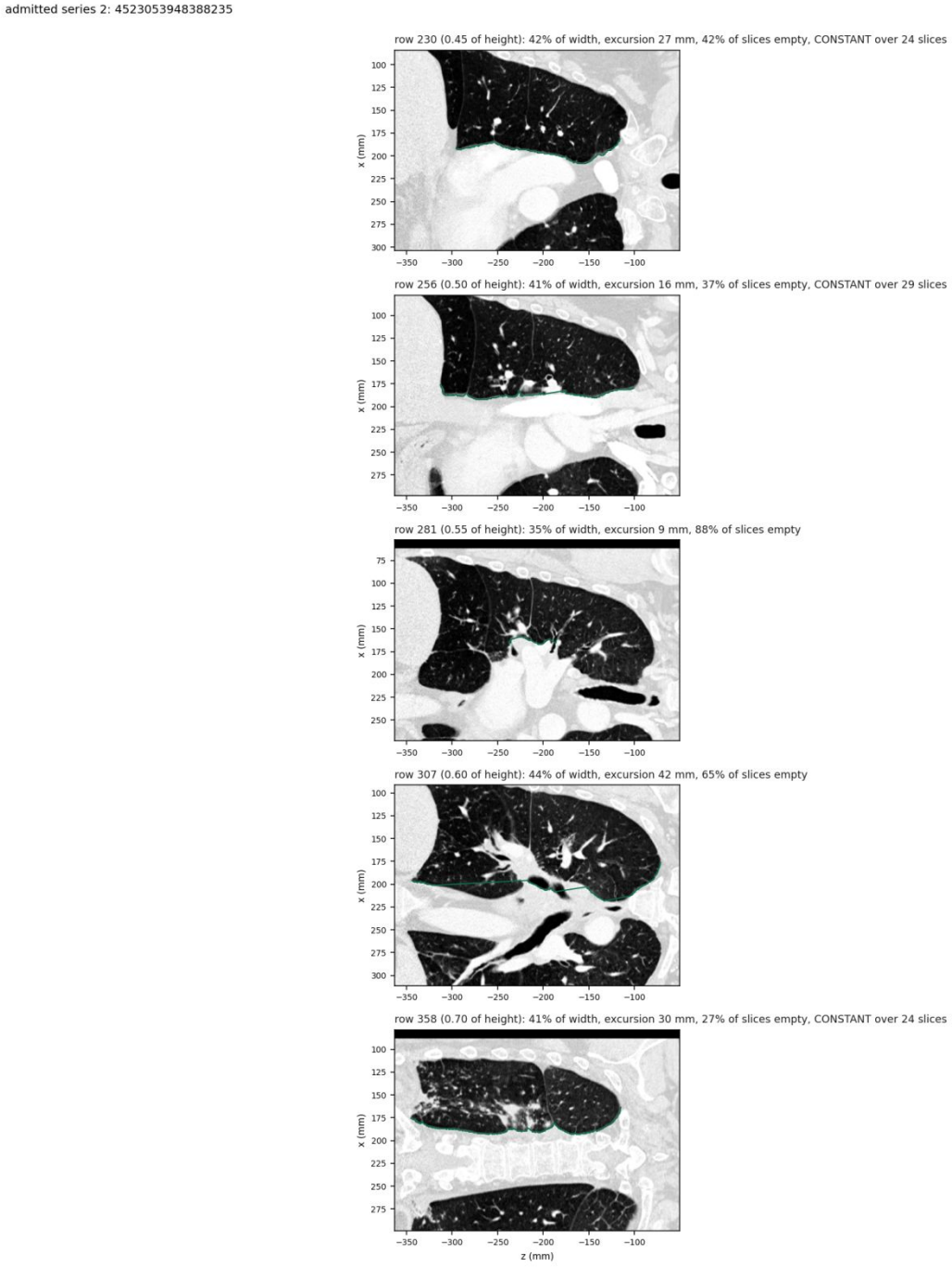


Figure S5. Admitted series 2, five usable levels.

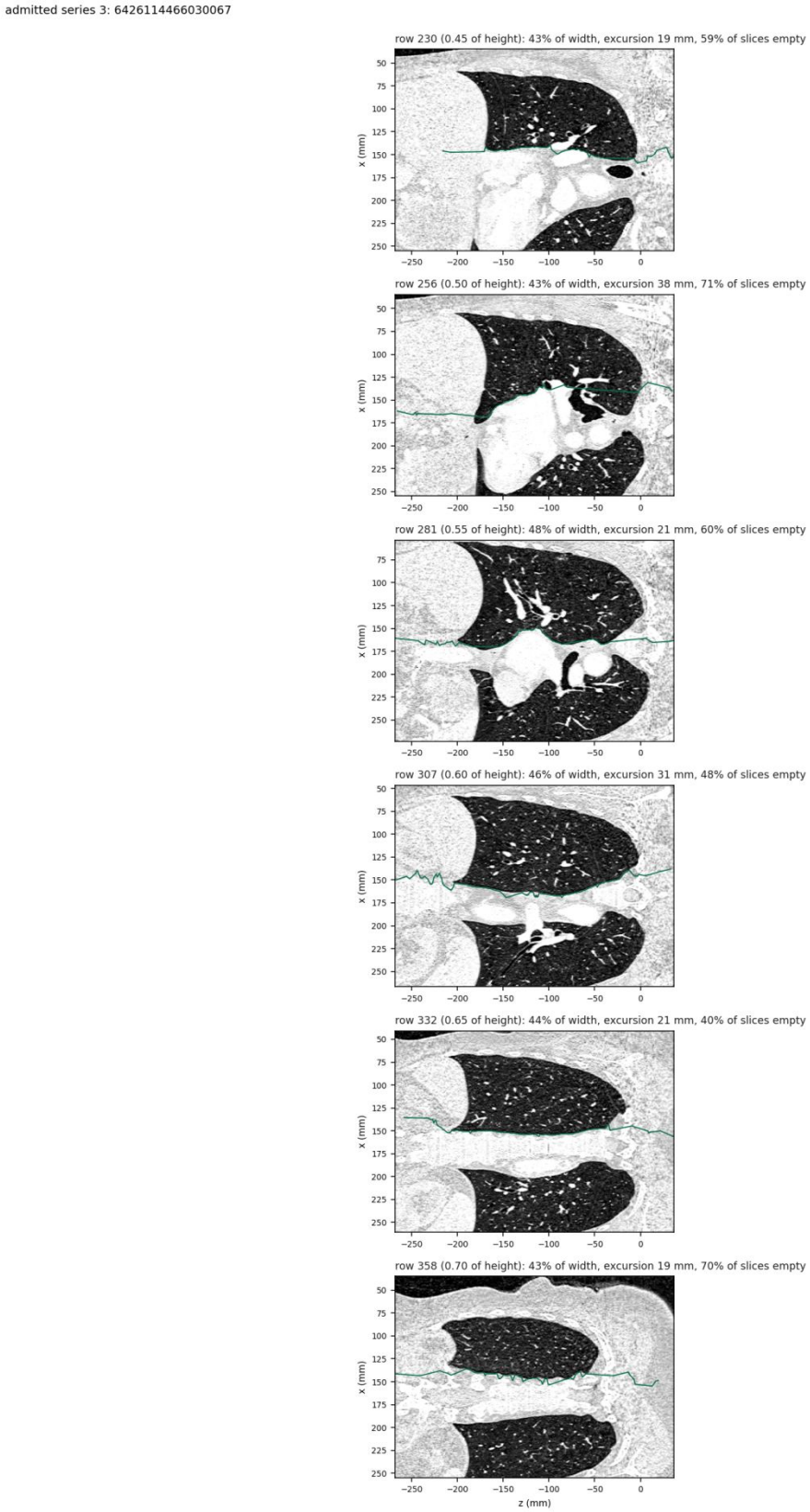


Figure S6. Admitted series 3, six usable levels. The tracked border continues past both ends of the lung, where the image is soft tissue on both sides of it; that is the series whose cycles fall from 2.77 to 1.68 when counted over the stretch where the border is at a lung interface.

Supplementary references

S1. Kendall A and Gal Y. What Uncertainties Do We Need in Bayesian Deep Learning for Computer Vision?. arXiv, https://doi.org/10.48550/arXiv.1703.04977 (2017, accessed 5 November 2026).

S2. Guo C, Pleiss G, Sun Y, et al. On Calibration of Modern Neural Networks. arXiv, https://doi.org/10.48550/arXiv.1706.04599 (2017, accessed 5 November 2026).

S3. Ovadia Y, Fertig E, Ren J, et al. Can You Trust Your Model's Uncertainty? Evaluating Predictive Uncertainty Under Dataset Shift. arXiv, https://doi.org/10.48550/arXiv.1906.02530 (2019, accessed 5 November 2026).

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